Homework 1

Tuesday, September 8, 2026 5:09 PM

1. $C \underset{b}{\overset{a}{\rightleftharpoons}} O \xrightarrow{c} I \xrightarrow{d} C$

a) $a = \alpha_{CO}$

$b = \alpha_{OC}$

$c = \alpha_{OI}$

$d = \alpha_{IC}$

Closed state C:

$C \xrightarrow{a} O$ $-aC$ → state loses channels

$O \xrightarrow{b} C$ $+bO$

$I \xrightarrow{d} C$ $+dI$ → state gains channels

$$\frac{dC}{dt} = -aC + bO + dI$$

Open state O:

$C \xrightarrow{a} O$ $+aC$

$O \xrightarrow{b} C$ $-bO$

$O \xrightarrow{c} I$ $-cO$

$$\frac{dO}{dt} = aC - (b+c)O$$

Inactive state I:

$O \xrightarrow{c} I$ $+cO$

$I \xrightarrow{d} C$ $-dC$

$$\frac{dI}{dt} = cO - dI$$

$$\boxed{\begin{aligned} \therefore C' &= -aC + bO + dI \\ O' &= aC - (b+c)O \\ I' &= cO - dI \end{aligned}}$$

$C' + O' + I' = -aC + bO + dI + aC - bO - cO + cO - dI$

$C' + O' + I' = 0$

$\boxed{\therefore C(t) + O(t) + I(t) = \text{constant and system is conserved}}$

Since C, O, I are fractions: $C + O + I = 1$

channels aren't being created or destroyed, if a channel departs one state, it enters another.

b) $C + O + I = 1$

$\Rightarrow I = 1 - C - O$

$C' = -aC + bO + d(1 - C - O)$

$$\boxed{\begin{aligned} \therefore C' &= d - (a+d)C + (b-d)O \\ O' &= aC - (b+c)O \end{aligned}}$$

$$J = \begin{bmatrix} C'_C & C'_O \\ O'_C & O'_O \end{bmatrix}$$

$$\boxed{\therefore J = \begin{bmatrix} -a-d & b-d \\ a & -b-c \end{bmatrix}}$$

$\text{tr}(J) = -a - b - c - d$

$= -(a+b+c+d)$

$\boxed{\text{all rates } a,b,c,d > 0 \quad \therefore \text{tr}(J) < 0}$

$\det(J) = (-a-d)(-b-c) - a(b-d)$

$= ab + ac + bd + cd - ab + ad$

$= ac + ad + bd + cd$

$\boxed{\text{all rates } a,b,c,d > 0 \quad \therefore \det(J) > 0}$

$\lambda_1 + \lambda_2 = \text{tr}\, J < 0$

and

$\lambda_1 \lambda_2 = \det J > 0$

So both have negative real parts.

$\boxed{\therefore \text{The equilibrium is stable}}$

c) $C' = O' = I' = 0$

$O' = aC - (b+c)O = 0$

$\bar{C} = \frac{b+c}{a}\bar{O}$

$C' = 0 = d - (a+d)C + (b-d)O$

$0 = d - (a+d)\frac{(b+c)}{a}O + (b-d)O$

$d = \left[\frac{(a+d)(b+c) + a(b-d)}{a}\right]O$

$d = \frac{ac + ad + bd + cd}{a}O$

$$\boxed{\bar{O} = \frac{ad}{ac + ad + bd + cd} = \frac{ad}{\det J}}$$

$\bar{C} = \frac{b+c}{a}\left(\frac{ad}{ac + ad + bd + cd}\right)$

$$\boxed{\bar{C} = \frac{d(b+c)}{ac + ad + bd + cd} = \frac{d(b+c)}{\det J}}$$

$I = 1 - C - O$

$= \frac{ac + ad + bd + cd - bd - cd - ad}{\det J}$

$$\boxed{\bar{I} = \frac{ac}{\det J}}$$

$$\boxed{\therefore (\bar{C}, \bar{O}, \bar{I}) = \frac{1}{\det J}\left(d(b+c), ad, ac\right)}$$

$I \xrightarrow{d} C$

As $d \to 0^+$ $\det J \to ac$

So $\bar{I} \to 1$

and thus $\bar{C} \to 0$ and $\bar{O} \to 0$.

This make sense because as $d \to 0^+$ the inactive channels take a long time to recover. $I \xrightarrow{\text{slowly}} C$

So $d \approx 0$: a very inactive state.

2. $u' = u - \frac{u^3}{3} - w + I$ $\epsilon > 0$

$w' = \epsilon(u - \gamma w)$ $\gamma > 0$

a) $u - \frac{u^3}{3} - w + I = 0$

$\boxed{w = u - \frac{u^3}{3} + I}$ → nullcline 1

$\epsilon(u - \gamma w) = 0$

$u - \gamma w = 0$

$\boxed{w = \frac{u}{\gamma}}$ → nullcline 2

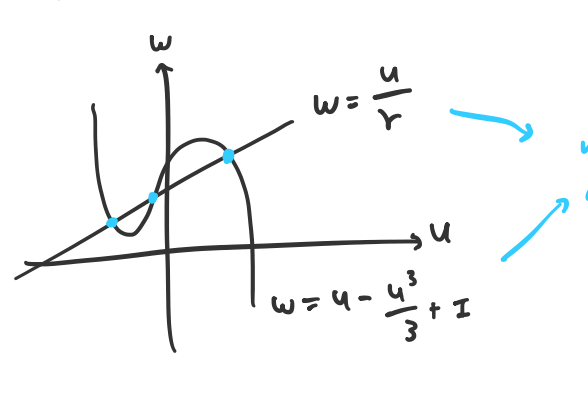
For equilibria:

$\frac{u}{\gamma} = u - \frac{u^3}{3} + I$

$$\boxed{\therefore \frac{u^3}{3} + \left(\frac{1}{\gamma} - 1\right)u = I}$$

$\left[\frac{u^3}{3} + \left(\frac{1}{\gamma} - 1\right)u\right]$ is a cubic polynomial which can have at most three real roots.

graphically:



let $F(u) = \frac{u^3}{3} + \left(\frac{1}{\gamma} - 1\right)u$

$F'(u) = u^2 + \frac{1}{\gamma} - 1$

if $\gamma < 1$

then $\frac{1}{\gamma} - 1 > 0$

and $F'(u) > 0$: $F(u)$ is strictly increasing

if $\gamma = 1$

then $F(u) = \frac{u^3}{3}$ is also strictly increasing

Thus $F(u) = I$ has exactly one u mapped to each I

$\boxed{\therefore \text{one equilibrium for } \gamma \leq 1.}$

$u' = u - \frac{u^3}{3} - w + I = f(u, w)$

$w' = \epsilon(u - \gamma w) = g(u, w)$

$$\boxed{\therefore J(\bar{u}, \bar{w}) = \begin{pmatrix} 1 - \bar{u}^2 & -1 \\ \epsilon & -\epsilon\gamma \end{pmatrix}}$$

b) Fix $\gamma = 1$:

$\frac{\bar{u}^3}{3} + \left(\frac{1}{\gamma} - 1\right)\bar{u} = I \xrightarrow{\gamma=1} \frac{\bar{u}^3}{3} = I$

$\bar{u} = (3I)^{1/3}$

$\bar{w} = \frac{\bar{u}}{\gamma} \xrightarrow{\gamma=1} \bar{w} = \bar{u}$

$\boxed{\therefore \bar{u} = \bar{w} = (3I)^{1/3}}$

$$J = \begin{pmatrix} 1 - \bar{u}^2 & -1 \\ \epsilon & -\epsilon\gamma \end{pmatrix}$$

$\det J = (1 - \bar{u}^2)(-\epsilon\gamma) + \epsilon$

$\det J = -\epsilon + \epsilon\bar{u}^2 + \epsilon$ → set $\gamma = 1$

$\boxed{\det J = \epsilon\bar{u}^2}$

For $I \neq 0$ then $\bar{u} \neq 0$

$\boxed{\therefore \det J > 0 \quad \text{since } \epsilon > 0}$

$\det J < 0$		saddle → unstable
$\det J > 0$	$\text{tr}\, J < 0$	stable
$\det J > 0$	$\text{tr}\, J > 0$	unstable
$\det J > 0$	$\text{tr}\, J = 0$	possible bifurcation

Since $\det J > 0$

then $\text{tr}\, J < 0$: stable

$\text{tr}\, J > 0$: unstable

$\text{tr}\, J = 1 - \bar{u}^2 - \epsilon\gamma$

$\text{tr}\, J = 1 - \bar{u}^2 - \epsilon$ → for $\gamma = 1$

$$\boxed{\begin{aligned} &\text{unstable if: } 1 - \bar{u}^2 - \epsilon > 0 \\ &\therefore \bar{u}^2 < 1 - \epsilon \end{aligned}}$$

Since $\bar{u}^2 \geq 0$ then only unstable

if $0 < 1 - \epsilon \to \epsilon < 1$

$\boxed{\therefore \text{repetitive spiking instability requires } \epsilon < 1}$

c) $I = \pm I_c$

$I_c = \frac{(1-\epsilon)^{3/2}}{3}$

bifurcation at $\text{tr}\, J = 0$ $\det J > 0$ already showed this

$\text{tr}\, J = 1 - \epsilon - \bar{u}^2 = 0$

$\bar{u}^2 = 1 - \epsilon$

$\bar{u} = \pm\sqrt{1 - \epsilon}$

at $\bar{u} = (3I)^{1/3}$:

$(3I)^{1/3} = \pm\sqrt{1 - \epsilon}$

$$\boxed{\therefore \text{bifurcation at: } I = \pm\frac{(1-\epsilon)^{3/2}}{3}}$$

recall $\text{tr}\, J > 0$: unstable

$\text{tr}\, J < 0$: stable

$\text{tr}\, J = 1 - \epsilon - \bar{u}^2$

$0 > 1 - \epsilon - |3I|^{2/3}$ → stable

$|I| > \frac{(1-\epsilon)^{3/2}}{3}$ → stable

$$\boxed{\begin{aligned} \therefore |I| &> I_c \quad \text{stable (rests)} \\ |I| &< I_c \quad \text{unstable (spikes)} \end{aligned}}$$

$I_c(\epsilon = \tfrac{3}{4}) = \frac{\left(1 - \frac{3}{4}\right)^{3/2}}{3}$

$= \frac{\left(\frac{1}{4}\right)^{3/2}}{3}$

$\boxed{\therefore I_c = \frac{1}{24}}$

3. $\frac{dv}{dt} = v^2 + I$

$v(T^+) \mapsto -\infty$ when $v(T^-) \to +\infty$

a) $\frac{dv}{dt} = v^2 + I$

$dt = \frac{dv}{v^2 + I}$

$\int dt = \int \frac{dv}{v^2 + I}$

note: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$

$t + C = \frac{1}{\sqrt{I}}\tan^{-1}\left(\frac{v}{\sqrt{I}}\right)$

$\frac{v}{\sqrt{I}} = \tan\left(\sqrt{I}t + C\right)$ → absorbs $\sqrt{I}$ into C

$v(t) = \sqrt{I}\tan\left(\sqrt{I}t + C\right)$

$v(0) = -\infty = \sqrt{I}\tan(C)$

$C = \tan^{-1}(-\infty) = \frac{-\pi}{2}$

$$\boxed{\therefore v(t) = \sqrt{I}\tan\left(\sqrt{I}t - \frac{\pi}{2}\right)}$$

$v(t)$ reaches $+\infty$ at $v(T) = \sqrt{I}\tan\left(\sqrt{I}T - \frac{\pi}{2}\right) = +\infty$

we need $\tan\left(\sqrt{I}T - \frac{\pi}{2}\right) = +\infty$

$\to \sqrt{I}T - \frac{\pi}{2} = \frac{\pi}{2}$

$\sqrt{I}T = \pi$

$$\boxed{\begin{aligned} \therefore T &= \frac{\pi}{\sqrt{I}} \\ f &= \frac{1}{T} = \frac{\sqrt{I}}{\pi} \end{aligned}}$$

b) $v = \tan\left(\frac{\theta}{2}\right)$

$\frac{dv}{dt} = \frac{1}{2}\sec^2\left(\frac{\theta}{2}\right)\frac{d\theta}{dt}$

$\frac{dv}{dt} = v^2 + I$

$\frac{1}{2}\sec^2\left(\frac{\theta}{2}\right)\frac{d\theta}{dt} = \tan^2\left(\frac{\theta}{2}\right) + I$

$\frac{1}{2}\frac{1}{\cos^2\left(\frac{\theta}{2}\right)}\frac{d\theta}{dt} - I = \frac{\sin^2\left(\frac{\theta}{2}\right)}{\cos^2\left(\frac{\theta}{2}\right)}$

$\frac{1}{2}\frac{d\theta}{dt} - I\cos^2\left(\frac{\theta}{2}\right) = \sin^2\left(\frac{\theta}{2}\right)$

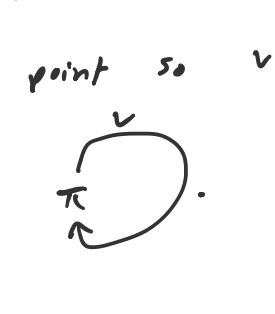
$\frac{d\theta}{dt} = 2\sin^2\left(\frac{\theta}{2}\right) + 2I\cos^2\left(\frac{\theta}{2}\right)$

$$\boxed{\therefore \frac{d\theta}{dt} = (1 - \cos\theta) + I(1 + \cos\theta)}$$

This form removes the need for a reset condition because it maps:

$-\infty \xrightarrow{v} +\infty$ to $-\pi \xrightarrow{v} +\pi$.

In the form of a circle $-\pi$ and π are the same point so $v(t)$ becomes a smooth cycle



c) For $I < 0$:

$\theta' = 0 = (1 - \cos\theta) + I(1 + \cos\theta)$

$\cos\theta - I\cos\theta = 1 + I$

$\cos\theta = \frac{1 + I}{1 - I}$

$\theta = \cos^{-1}\left(\frac{1 + I}{1 - I}\right)$

$$\boxed{\text{on } [-\pi, \pi] \text{ that means: } \theta = \pm\cos^{-1}\left(\frac{1 + I}{1 - I}\right)}$$

let $\theta' = F(\theta) = (1 - \cos\theta) + I(1 + \cos\theta)$

$J = [F'(\bar{\theta})]$ → for single ODE

$F'(\theta) = \sin\theta - I\sin\theta$

$\lambda = (1 - I)\sin\bar{\theta}$

plug into equilibrium equation:

$\bar{\theta} = \pm\cos^{-1}\left(\frac{1 + I}{1 - I}\right)$

use $\sin(\pm\cos^{-1}x) = \pm\sqrt{1 - x^2}$

$\sin\bar{\theta} = \pm\sqrt{1 - \left(\frac{1 + I}{1 - I}\right)^2}$

$\frac{\lambda}{(1 - I)} = \pm\sqrt{\frac{(1 - I)^2 - (1 + I)^2}{(1 - I)^2}}$

$\lambda = \pm(1 - I)\sqrt{\frac{1 - 2I + I^2 - 1 - 2I - I^2}{(1 - I)^2}}$

$\lambda = \pm(1 - I)\sqrt{\frac{-4I}{(1 - I)^2}}$

Since $I < 0$ we have $-I = |I|$

Also $(1 - I) > 0$ so $\sqrt{(1 - I)^2} = 1 - I$

$\lambda = \pm(1 - I)\frac{2\sqrt{|I|}}{1 - I}$

$\boxed{\therefore \lambda = \pm 2\sqrt{|I|}}$

$\lambda = 2\sqrt{|I|} > 0$: unstable

$\lambda = -2\sqrt{|I|} < 0$: stable

As $I \to 0^-$ the eigenvalues approach eachother at zero, where they annihilate eachother.

A $I = 0$, there is one equilibrium $\frac{dv}{dt} = 0 \Rightarrow v^2 = -I \xrightarrow{I=0} v = 0$

$\xrightarrow{I>0}$ no real v

and at $I > 0$ there are no equilibria.

This is a saddle-node bifurcation.

In a) we saw $f = \frac{\sqrt{I}}{\pi}$: as $I \to 0^+$ $f \to 0$

we see $I < 0$ rest and $I > 0$ repetitive firing

In a real neuron this implies that above the threshold it can fire at arbitrarily low frequencies.

4. $\tau_m\frac{du}{dt} = -\left[u(t) - u_{rest}\right] + RI(t)$

$I(t) = I_0 e^{-t/\tau_s}$ $u(0) = 0$ $u_{rest} = 0$

a) $\tau_m u' = -u + RI_0 e^{-t/\tau_s}$

$u' + \frac{1}{\tau_m}u = KI_0 e^{-t/\tau_s}$

$\mu(t) = e^{\int \frac{1}{\tau_m}dt}$

$= e^{t/\tau_m}$

$\frac{d}{dt}\left(e^{t/\tau_m}u\right) = \frac{RI_0}{\tau_m}e^{t/\tau_m - t/\tau_s}$

$e^{t/\tau_m}u(t) - u(0) = \frac{RI_0}{\tau_m}\int_0^t e^{s\left(\frac{1}{\tau_m} - \frac{1}{\tau_s}\right)}ds$

$e^{t/\tau_m}u(t) = \frac{RI_0}{\tau_m}\frac{e^{t\left(1/\tau_m - \frac{1}{\tau_s}\right)} - 1}{\frac{1}{\tau_m} - \frac{1}{\tau_s}}$

$u(t) = e^{-t/\tau_m}\frac{RI_0\tau_s}{\tau_s - \tau_m}\left(e^{t\left(1/\tau_m - \frac{1}{\tau_s}\right)} - 1\right)$

$$\boxed{\therefore u(t) = \frac{RI_0\tau_s}{\tau_s - \tau_m}\left[e^{-t/\tau_s} - e^{-t/\tau_m}\right]}$$

b) let $\frac{RI_0\tau_s}{\tau_s - \tau_m} = A$

$u'(t) = A\left[\frac{-1}{\tau_s}e^{-t/\tau_s} + \frac{1}{\tau_m}e^{-t/\tau_m}\right] = 0$

$\frac{1}{\tau_m}e^{-t/\tau_m} = \frac{1}{\tau_s}e^{-t/\tau_s}$

$\frac{\tau_s}{\tau_m} = e^{t\left(\frac{1}{\tau_m} - \frac{1}{\tau_s}\right)}$

$\log\left(\frac{\tau_s}{\tau_m}\right) = t\left(\frac{1}{\tau_m} - \frac{1}{\tau_s}\right) = t\left(\frac{\tau_s - \tau_m}{\tau_m\tau_s}\right)$

$$\boxed{\therefore t_{peak} = \frac{\tau_m\tau_s}{\tau_s - \tau_m}\log\left(\frac{\tau_s}{\tau_m}\right)}$$

$r = \frac{\tau_s}{\tau_m}$

$\frac{t}{\tau_s} = \frac{1}{\frac{\tau_s}{\tau_m} - 1}\log r$

$\frac{t}{\tau_s} = \frac{1}{r - 1}\log r$

$e^{-t/\tau_s} = e^{\frac{-\log r}{r-1}} = r^{\frac{-1}{r-1}}$

alternatively:

$\frac{t}{\tau_m} = \frac{1}{1 - \frac{\tau_m}{\tau_s}}\log r = \frac{1}{\frac{r-1}{r}}\log r$

$\frac{t}{\tau_m} = \frac{r}{r-1}\log r$

$e^{-t/\tau_m} = e^{\frac{r}{r-1}\log r} = r^{\frac{-r}{r-1}}$

Then:

$u(t) = \frac{RI_0\tau_s}{\tau_s - \tau_m}\left[e^{-t/\tau_s} - e^{-t/\tau_m}\right]$

$u = RI_0\frac{1}{\frac{r-1}{r}}\left(r^{\frac{-1}{r-1}} - r^{\frac{-r}{r-1}}\right)$

$u = RI_0\frac{r}{r-1}\left(r^{\frac{-1}{r-1}}\right)\left(1 - \frac{1}{r}\right)$

$u = RI_0\frac{r}{r-1}\left(r^{\frac{-1}{r-1}}\right)\left(\frac{r-1}{r}\right)$

$\boxed{\therefore u_{max} = RI_0 r^{-1/r-1}}$!

Find I where $u_{max} = u_{th}$

$RI_0 r^{-1/r-1} = u_{th}$

$I_0^{crit} = \frac{u_{th}}{R}r^{1/r-1}$

$I_0^{crit} = \frac{u_{th} - u_{rest}}{R}r^{1/r-1}$

$r^{1/r-1} = e^{\frac{\log r}{r-1}}$

as $r = \frac{\tau_s}{\tau_m} \to \infty$: $\frac{\log r}{r-1} \to 0$

so $e^{\log r/r-1} \to e^0 = 1$

$$\boxed{\therefore I_0^{crit} \to \frac{u_{th} - u_{rest}}{R}}$$